higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

11 December 2025 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 6 pages and a formula sheet of 2 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Write neatly and legibly.

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SECTION A

QUESTION 1: FLOW MEASUREMENT

1.1 The orifice meter is a common and simple method to measure flowrate. It consists of a thin, steel plate with a hole through which the flow passes.

1.1.1 Describe the construction of a segmental orifice plate. (2)

1.1.2 Explain why this orifice is used where dirty fluids are encountered. (1)

1.1.3 Name THREE other differential pressure primary devices. (3)

1.2 What is a *weir*? (2)

1.3 Instruments used in the measurement of flow can be divided into two main classes; namely rate of flow meters and volumetric meters:

1.3.1 Explain the operational differences between these classes of meters. (2 × 2) (4)

1.3.2 Each instrument may be considered as two parts, the primary element and the secondary element. Define the two elements. (2)

1.3.3 What is the purpose of measuring flow elements? (1)

[15]

QUESTION 2

There are basically two types of thermal mass flow meters: those that measure temperature rise of fluid, and those that measure the rate of heat loss to the fluid stream.

2.1 Make a neat, labelled sketch of a thermal mass flow meter that measures temperature rise of a fluid. (5)

2.2 Explain the working principle of this heat transfer flow meter. (7)

2.3 What is the disadvantage of this flow meter? (2)

2.4 What is the advantage of the rate of heat loss flow meter over the heat transfer flow meter? (1)

[15]

TOTAL SECTION A: 30

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SECTION B

QUESTION 3: DENSITY, HUMIDITY AND VISCOSITY MEASUREMENT

3.1 Piezo-electric quartz crystals have several uses in the industry.

3.1.1 Name FIVE uses of piezo-electric quartz crystals. (5)

3.1.2 Briefly explain the principle of operation of a piezo electric hygrometer. (7)

3.2 Name FIVE points which should be taken into account when using a float viscometer. (5)

3.3 One of the methods of measuring specific gravity and density is by utilising a balance flow vessel method:

3.3.1 Make a neatly labelled sketch that matches the above description. (5)

3.3.2 Explain the operation of the instrument in QUESTION 3.3.1. (5)

3.4 FIGURE 1 below is a sketch of a viscometer designed for continuous measurement of viscosity in process streams.

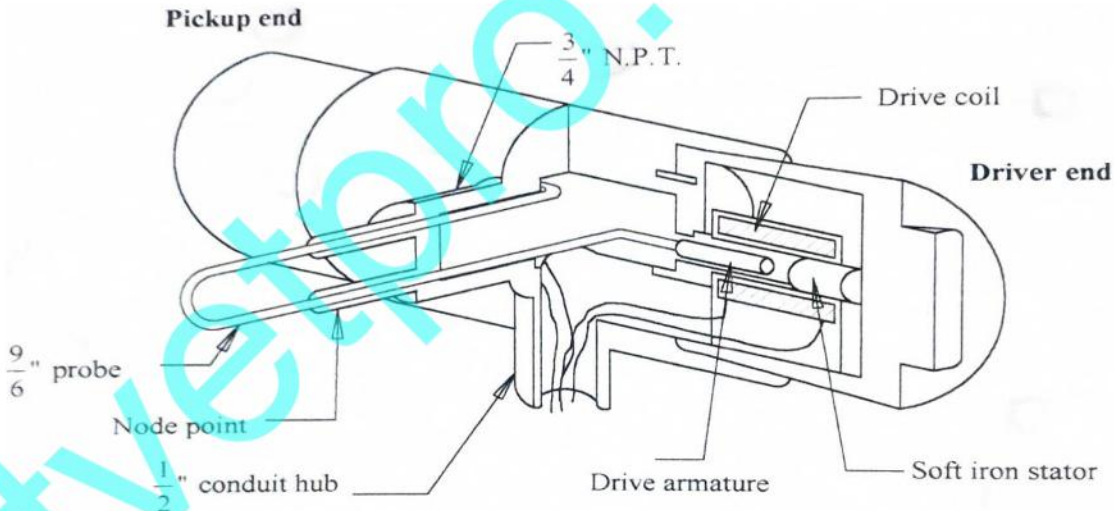


FIGURE 1

3.4.1 Name the viscometer shown in FIGURE 1 above. (1)

3.4.2 Explain the working principle of the drive end of this viscometer. (5)

[33]

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QUESTION 4: PH AND CONDUCTIVITY MEASUREMENT

4.1 What is the main purpose of a reference electrode? (2)

4.2 FIGURE 2 below is a sketch of a glass vessel containing a solution of hydrochloric acid.

Study it and answer the following questions:

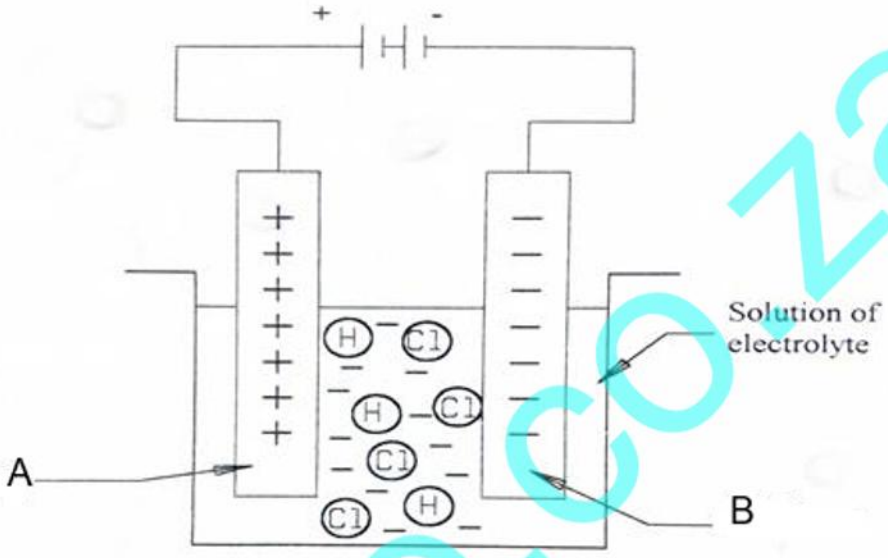


FIGURE 2

4.2.1 Name the electrode marked A.

4.2.2 Name the electrode marked B.

4.2.3 Name the anions in the solution.

4.2.4 Name the cations in the solution.

(4 × 1) (4)

4.3 Define the term, *electrolysis*. (3)

4.4 What is *polarization*? (3)

4.5 Define the pH of a solution. (2)

[14]

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QUESTION 5: AUTOMATIC - CONTROL

5.1 Name THREE functional elements needed to perform the functions of automatic systems. (3)

5.2 Give THREE practical applications where ON/OFF controller action is used. (3)

5.3 Industrial processes are characterised by the many types of control variables.

5.3.1 Name any FOUR control variables that are controlled in industrial processes. (4)

5.3.2 Name FIVE forms of control modes that are commonly used in the industry. (5)

5.4 The important time delays can be classified under two general categories; namely distance velocity lag and transfer lags.

5.4.1 Define *distance velocity lag*. (2)

5.4.2 Make a neat, labelled sketch showing distance velocity lag in a control loop. The control loop must show a process and the detecting element. (4)

5.4.3 Name TWO factors that influences transmission lags. (2)
[23]

TOTAL SECTION B: 70
GRAND TOTAL: 100

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INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$W = 359 \cong 2\ CZ,Ed^2\sqrt{(h\rho)}$ $\quad R_d = W/15 \cong 8\ \mu d$

$Q = 359 \cong 2\ CZ,Ed^2\sqrt{(h/\rho)}$ $\quad R_d = Q\Delta/15 \cong 8\ \mu d$

$W = 0 \cong 01252\ CZ,Ed^2\sqrt{(h\rho)}$ $\quad R_d = 3 \cong 54\ W/\mu d$

$Q = 0 \cong 01252\ CZ,Ed^2\sqrt{(h/\rho)}$ $\quad R_d = 3 \cong 54\ Q\Delta/\mu d$

$Q_g = 2238\ CZEd^2\sqrt{(h/\rho)}$ $\quad R_d = Q_g\Delta/98 \cong 6\ \mu d$

$m = (d/D)^2$ $\quad E = 1/\sqrt{(1-m^2)}$

$$N = \frac{W}{0 \cdot 01252 D^2\sqrt{(h\rho)}} = \frac{Q\sqrt{(\rho)}}{0 \cdot 01252\ D^2\sqrt{(h)}}$$

$mE = N/CZ,$ $\quad CmE = N/Z,$ $\quad mE = CmE/C$

$m = (d/D)^2$ $\quad E = 1/\sqrt{(1-m^2)}$

$$R_d = \frac{W}{15 \cdot 8\ \mu D\sqrt{(m)}} = \frac{Q\rho}{15 \cdot 8\ \mu D\sqrt{(m)}} = \frac{Q_g\rho}{98 \cdot 6\ \mu D\sqrt{(m)}}$$

$$N = \frac{W}{359 \cdot 2\ D^2\sqrt{(h)}} = \frac{Q\sqrt{(\rho)}}{359 \cdot 2\ D^2\sqrt{(h)}} = \frac{Q_g\sqrt{(\rho)}}{2238\ D^2\sqrt{(h)}}$$

$d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$ $\quad mE = N/CZ,$

$W = 1890\ Ud^2\sqrt{(\rho P)}$ *for critical flow/ vir kritieke vloei*

$CmE = N/Z,$ $\quad mE = CmE/C$

$d = [W/1\ 890U\sqrt{(\rho P)}]^{\frac{1}{2}}$ *for critical flow/vir kritieke vloei*

$$R_d = \frac{3 \cdot 54W}{\mu D\sqrt{(m)}} = \frac{3 \cdot 54Q\rho}{\mu D\sqrt{(m)}}$$ $\quad d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$

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$W = 1{\cdot}252\,Ud^2\sqrt{(\rho P)}$ *for critical flow/ vir kritieke vloei*

$d = [W/1{\cdot}252\,U\sqrt{(\rho P)}]^{\frac{1}{2}}$ *for critical flow/vir kritieke vloei*

1 kPa = 102 mmWD = 102 mmWG

1 lb/ft³ = 16,0183 kg/m³

Atmospheric pressure/Atmosferiese druk = 101,325 kPa

Gravitation acceleration/Gravitasieversnelling = 9,81 m/s²

For/ Vir D + D/2 tappings/tappe and/en flange tappings/flenstappe

$$\frac{h}{Pa} \times 27{,}2 = \frac{kPa}{kPa} \times 27{,}2$$

$$Q = \frac{8}{15} \quad Tan\,\frac{\theta}{2}\sqrt{2g.H^5} \qquad Q = \frac{2}{3} \quad B\sqrt{2g.H^3}$$

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